

1) student management system:

START

class student

Private name, rollNo, marks1, marks2, marks3, average

Method acceptDetails()

Input name, rollNo, marks1,

End method

Method result()

If marks ≥ 50

Display "Pass"

else

Display "Fail"

End if

End method

Main

CREATE student s

s.accept()

s.result()

2) Library Book Management:

class Book

Private title, author, ISBN, available

Method issue()

available = false

End method

```
METHOD returnBook()
```

```
    available = TRUE
```

```
END METHOD
```

```
METHOD display()
```

```
    DISPLAY title, author, ISBN, available
```

```
END METHOD
```

```
MAIN
```

```
CREATE Book b1, b2, b3
```

```
    b1.issue()
```

```
    b1.returnBook()
```

```
    b1.display
```

3) Employee salary calculation

```
CLASS Employee
```

```
PRIVATE empId, name, basicPay,
```

```
METHOD grossSalary()
```

```
    gross = basicPay + HRA + DA
```

```
END METHOD
```

```
METHOD netSalary()
```

```
    net = gross - PF - Tax.
```

```
END METHOD.
```

```
METHOD. display()
```

```
    DISPLAY gross, net.
```

```
END METHOD.
```

```
MAIN
```

```
CREATE employee e.
```

```
    e.grossSalary()
```

```
    e.netSalary()
```

```
    e.display()
```

4) Shape area calculator

CLASS shape

METHOD areac)

END METHOD

CLASS circle EXTENDS shape.

METHOD areac)

Display $3.14 * r * r$

END METHOD

CLASS Rectangle EXTENDS shape.

METHOD areac)

Display $l * b$

END METHOD

CLASS Triangle EXTENDS shape.

METHOD areac)

Display $0.5 * base * height$

END METHOD

5) Vehicle Registration system

CLASS vehicle

regNo, owner

METHOD display()

Display regNo, owner

END METHOD.

CLASS Two Wheeler EXTENDS vehicle

METHOD display()

Display "Bike Details"

END METHOD.

CLASS Four Wheeler EXTENDS vehicle

METHOD display()

Display "Car Details"

6) Bank

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7) H

6) Bank account hierarchy:

```
CLASS BANKAccount  
balance
```

```
METHOD withdraw()
```

```
END METHOD
```

```
CLASS SavingsAccount EXTENDS BANKAccount
```

```
METHOD withdraw()
```

```
    DISPLAY "withdrow from saving"
```

```
END METHOD
```

```
CLASS CurrentAccount EXTENDS BANKAccount
```

```
METHOD withdraw()
```

```
    DISPLAY "withdraw with over draft"
```

```
END METHOD
```

7) Hospital patient Record system:

```
CLASS Patient
```

```
METHOD generateBill()
```

```
END METHOD
```

```
CLASS Inpatient EXTENDS Patient
```

```
METHOD generateBill()
```

```
    DISPLAY "Room charges"
```

```
END METHOD
```

```
CLASS Outpatient EXTENDS Patient
```

```
METHOD generationBill()
```

```
    DISPLAY "consultation fee"
```

```
END METHOD
```

8) E-commerce Product catalog:

CLASS Product

METHOD applyDiscount()

END METHOD

CLASS Electronics EXTENDS Product

METHOD applyDiscount()

DISPLAY "10% Discount"

END METHOD

CLASS Clothing EXTENDS Product

METHOD applyDiscount()

DISPLAY "20% Discount"

END METHOD

CLASS Grocery EXTENDS Product

METHOD applyDiscount()

DISPLAY "5% Discount"

END METHOD

9) University staff hierarchy:

CLASS staff

METHOD allowance()

END METHOD

CLASS Teachingstaff EXTENDS staff

METHOD allowance()

DISPLAY "Teaching Allowance"

END METHOD

CLASS Professor EXTENDS Teachingstaff

METHOD allowance()

DISPLAY "Professor Allowance"

END METHOD

CLASS lecture EXTENDS Teaching Staff.

METHOD allowance()

DISPLAY "Lecturer allowance"

END METHOD

10) Smart Home Device Controller

CLASS smartDevice

METHOD execute()

END METHOD

CLASS smartLight EXTENDS smartDevice.

METHOD execute()

DISPLAY "Light ON"

END METHOD

CLASS smartThermosstat EXTENDS smartDevice

METHOD execute()

Display "Temperature set".

END METHOD

CLASS smartlock EXTENDS smartDevice.

METHOD execute()

Display "Door locked"

END METHOD

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10 ✓